

IoT in Agriculture

Learning Objectives

- Analyze how IoT technology addresses challenges in modern agriculture
- Evaluate the economic and environmental benefits of precision farming
- Design an IoT solution for a specific agricultural problem

Engage (5-7 minutes)

Present a contrast: traditional farming (farmers checking crops by walking fields) versus precision agriculture (drones flying over fields, sensors in soil, automated irrigation). Show data: precision farming can reduce water usage by 30% and increase yields by 20%. Ask students: why isn't everyone using this technology?

Explore (10-12 minutes)

Students receive a case study of a farm facing challenges: drought, pest infestation, and labor shortage. They design an IoT solution using available sensors (soil moisture, weather stations, cameras, GPS trackers on equipment), connectivity options (LoRa for wide areas, cellular for real-time data), and automated responses (smart irrigation, pest alerts, autonomous tractors). They must justify cost versus benefit.

Explain (10-12 minutes)

IoT in agriculture enables precision farming—using data to optimize every aspect of crop production. Soil sensors measure moisture, pH, and nutrient levels in real time. Weather stations provide microclimate data for individual fields. Drones with multispectral cameras detect crop stress before it's visible. Automated irrigation systems water only where needed. Livestock wearables monitor health and location. GPS-guided tractors plant seeds at optimal spacing. These technologies reduce waste, increase yields, and enable farming with less labor.

Elaborate (8-10 minutes)

Case studies: how Israeli drip irrigation systems use IoT to farm in deserts, how Indian farmers use low-cost sensor networks to combat drought, and how vertical farms in Singapore use IoT to grow food in urban spaces. Discuss barriers to adoption in developing countries: cost, connectivity, digital literacy, and infrastructure. Examine the environmental impact: reduced pesticide use, water conservation, and lower carbon emissions from optimized operations.

Evaluate (5-8 minutes)

Students research an agricultural IoT project and write a feasibility report: what technology is used, what data it collects, how decisions are made, what the ROI is, and what challenges exist. Then they design their own IoT solution for a local agricultural problem (e.g., community garden, school farm). Assessment: technical accuracy, practical feasibility, and consideration of economic viability.